Rodrigo Pimenta

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Experience

Systems Engineer III — Nov 2021 – Present
Hughes Network Systems — *Germantown, MD*

- Second-highest committer of ∼15 engineers on the satellite gateway automation platform: 400+ commits across 53 delivered tickets spanning monitoring, alerting, self-healing, and reporting for five ground-segment platforms
- Root-caused and eliminated a platform-wide logging defect that silently discarded log records, cutting the log footprint from 19.68 GB to 58 MB and reclaiming ∼25 GB in production; built audit tooling that swept all 146 scheduled jobs for the same pattern instead of fixing them individually
- Root-caused two production failures: a 738 MB core dump traced with gdb to an unguarded deserialization path in a vendor gateway component, and a 177-process leak traced to a library deadlock at interpreter teardown; delivered both as actionable defect reports and shipped the supervision module that closed the leak
- Root-caused a manager-daemon memory leak across 8 storage clusters (∼0.8 GB/month growth) by isolating the single non-leaking cluster as a natural experiment, then drove remediation with engineering to prevent the storage-exhaustion path behind customer-facing outages
- Designed the platform's shared logging entry point, now used by every script and device module; introduced continuous integration to the organization; authored the 105-table database contract documentation; and migrated the action-audit data model with 26,857 rows reconciled exactly
- Accelerated deployment of 20 data center racks from four weeks to two using Ansible, cut hardware test validation effort by 50%, and automated L2/L3 network configuration to reduce configuration errors by 90% across 1,000+ satellite sites
- Led Tier 3 troubleshooting and 24/7 on-call rotation for ground system incidents, maintaining a 99.9% availability SLA; drove data-driven QoS traffic policy design (Tableau/SQL) reducing peak-time congestion by 18% and customer complaint calls by 24%

Technology Analyst — Jun 2018 – Dec 2019
Bank of America — *Charlotte, NC*

- Developed PowerShell and Python scripts to automate software validation and hardware testing across 20+ ATM models in the enterprise lab environment
- Built a reusable CI/CD deployment framework in Jenkins that standardized release processes and moved builds from weekly to hourly, reducing environment setup time by 15%

Mass Communication Specialist, E-3 (Active Duty) — Jan 2020 – Jan 2021
United States Navy Reserve — *Great Lakes, IL / Fort Meade, MD*

- Completed basic training and the Defense Information School (DINFOS) at Fort Meade, MD; honorably discharged

Systems Built

Ceph Storage Self-Healing Monitor | *Python, uv, Fabric/SSH, PostgreSQL, Grafana, pytest*

- Sole author (77 of 79 commits, 6,586 lines). Monitors 24 storage nodes across 8 clusters on a 3-minute cycle. Automated metadata-server failover gated by a four-verdict standby-capacity assessment so it never fails over into a worse node, with outcomes verified against the cluster log to distinguish a real failover from a same-node reclaim. Append-only event history across 7 PostgreSQL tables with additive-only schema migration, a 19-panel Grafana dashboard, and a pytest suite in CI

Config Integrity Self-Heal | *Python, asyncio, SFTP, PostgreSQL*

- Sole author (30 of 30 commits, 3,594 lines). Detects configuration drift and corruption across the gateway fleet over SFTP and restarts the affected service, modernizing a blocking-I/O workload onto asyncio with semaphore-bounded fan-out rather than process pools

Orbital Watch | *Python, Cloudflare Workers, CI/CD, pytest* | orbital.rodrigopimenta.com

- Deployed satellite pass-prediction and space-weather dashboard built on the thesis that a system should report its own degradation: each upstream source is isolated, falls back cache → committed snapshot, and is classified fresh/stale/failed on the page. 142 tests, offline-deterministic build, scheduled rebuild on a Cloudflare Worker cron with a dead-man check

Education

University of Maryland, A. James Clark School of Engineering — College Park, MD
Bachelor of Science in Computer Engineering | College Park Scholars — *Aug 2014 – Dec 2018*

Technical Skills

Programming: Python, SQL, Bash, PowerShell, JavaScript
Automation & CI/CD: Ansible, Jenkins, Bitbucket Pipelines, Git, uv, ruff, pre-commit, pytest, Fabric, Netmiko
Cloud, Data & Observability: Google Cloud (BigQuery), PostgreSQL, Grafana, Tableau, pandas, SQLAlchemy, psycopg2
Infrastructure & Ops: Linux/RHEL, Ceph, asyncio, gdb / core-dump analysis, logrotate, VMware, 24/7 On-Call
AI-Assisted Development: Kiro, Cursor, Claude Code
Languages: English (Fluent), Portuguese (Fluent), Spanish (Proficient)